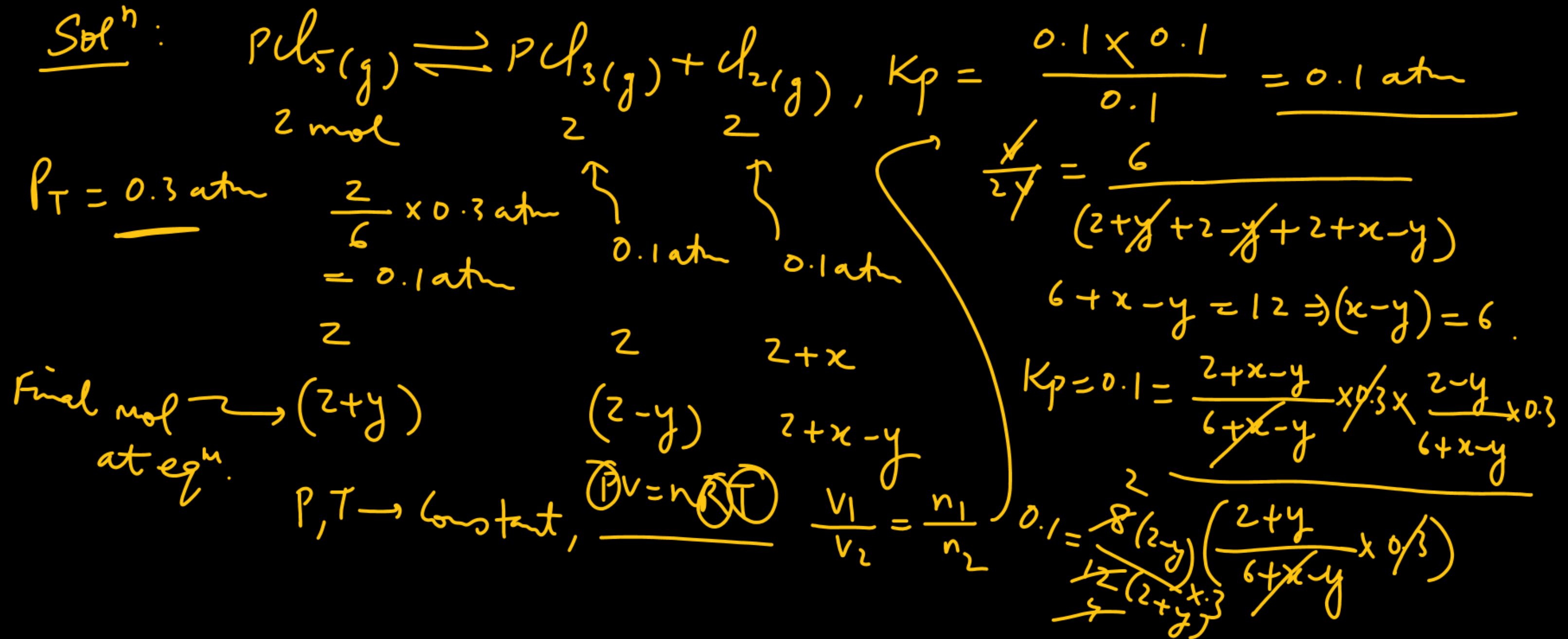


A container of capacity V L contains an equilibrium mixture that consists of 2 moles each of PCl_5 , PCl_3 and Cl_2 (all are gases). The pressure is 0.3 atm and temperature is T K. x mole of $\text{Cl}_2(\text{g})$ is now introduced keeping the pressure and temperature constant, until the equilibrium volume becomes $2V$ L. The value of x is :



$$2 + y = 4 - 2y$$

$$3y = 2$$

$$y = \frac{2}{3} \text{ mol.}$$

$$x - y = 6$$

$$x = 6 + \frac{2}{3}$$

$$= \frac{20}{3} \text{ mol} \quad \underline{\text{Ans}}$$

For the equilibrium $\text{NiO(s)} + \text{CO(g)} \rightleftharpoons \text{Ni(s)} + \text{CO}_2\text{(g)}$, $\Delta G^\circ(\text{cal/mol}) = -5320 - 5.6T$ (K). The temperature (in Kelvin) at which the gaseous mixture at equilibrium contains 400 ppm of CO by mole is [$\ln 10 = 2.3$, $\ln 2 = 0.7$]

Solⁿ: $\Delta G^\circ = -5320 - 5.6T$, $T = ?$,

$$-RT \ln K_p = -5320 - 5.6T$$

$$-2 \times T \times \ln \left(\frac{p_{\text{CO}_2}}{p_{\text{CO}}} \right) = -5320 - 5.6T$$

$$-2T \ln \frac{10^6}{400} = -5320 - 5.6T$$

$$-2T \left[\ln 10^4 - \ln 4 \right] = -5320 - 5.6T$$

$$-2T \left[4 \times 2.3 - 2 \times 0.7 \right] = -5320 - 5.6T$$

$$-15.6T = -5320 - 5.6T$$

$$T = 532 \text{ K}$$

$$n_T = 10^6 \text{ mol}$$

$$n_{\text{CO}} = 400 \text{ mol}$$

$$n_{\text{CO}_2} = 10^6 - 400 \times$$

$$\approx 10^6 \text{ mol}$$

$$\frac{9.2}{1.4}$$

$$\hline 7.8$$

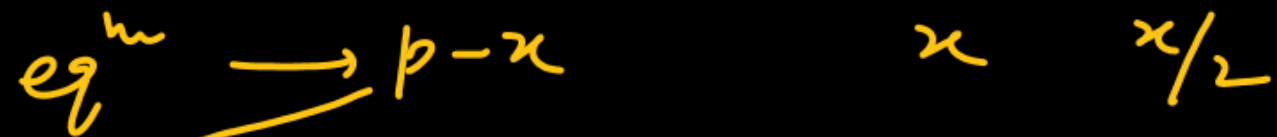
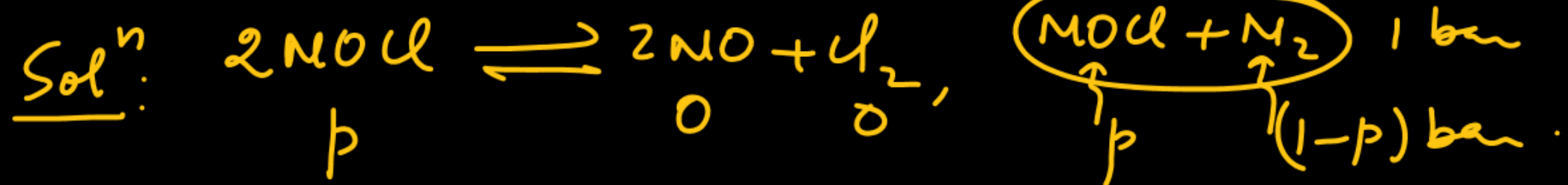
$$R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$$

$$= 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$\frac{n_{\text{CO}_2}}{n_{\text{CO}}} = \frac{p_{\text{CO}_2}}{p_{\text{CO}}}$$

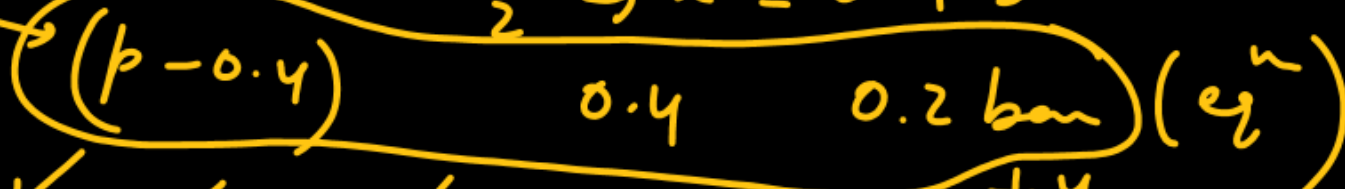
Ans

Gaseous nitrosyl chloride (NOCl) and N_2 are taken in a flask, sealed and heated to some temperature where the total pressure would have been 1.0 bar had not the following equilibrium been established $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$ But the actual pressure was found to be 1.2 bar. Now, into the equilibrium mixture, some Cl_2 gas was introduced at constant volume and temperature so that the total pressure would have been 8.3 bar had no further reaction occurred but the actual pressure was found to be 8.2 bar. The equilibrium constant K_P (in bar for the decomposition reaction under the given experimental condition is :



$$p_T = 1.2 = (p-x) + x + \frac{x}{2} + 1-p$$

$$1.2 = 1 + \frac{x}{2} \Rightarrow x = 0.4 \text{ bar}$$



$$p-0.4 + 0.4 + 0.2 + y + 1-p = 8.3$$

$$y = 7.1 \text{ bar}$$



$$\begin{matrix} p-0.4 & 0.4 \\ \uparrow +2z & \uparrow -2z \end{matrix}$$

$$p-0.4+2z + 0.4-2z + 7.3-z + 1-p$$

$$z = 0.1 \text{ bar} = 8.2$$

$(p-0.2) \quad 0.2 \text{ bar} \quad 7.2 \text{ bar}$

$$\frac{p-0.4}{p-0.2} = \frac{1}{3}$$

$$3p-1.2 = p-0.2$$

$$2p = 1$$

$$p = 0.5$$

$$K_P = \frac{0.4 \times 0.2}{(p-0.4)^2}$$

$$= \frac{0.2^2 \times 7.2}{(p-0.2)^2}$$

$$\frac{0.16 \times 0.2}{(p-0.4)^2} = \frac{0.04 \times 7.2}{(p-0.2)^2}$$

$$\left(\frac{p-0.4}{p-0.2}\right)^2 = \frac{8}{9}$$

$$\begin{aligned} K_p &= \frac{0.16 \times 0.2}{0.1 \times 0.1} \\ &= 16 \times 0.2 \\ &= 3.2 \text{ bar} \end{aligned}$$

Ans

Imp

For the coagulation of a particular colloidal solution, the order of coagulation power of some electrolytes is in the order $\text{Na}_3\text{PO}_4 > \text{CaSO}_4 > \text{AlCl}_3$. Which of the following information(s) may be correct for such colloid?

(A) In electro-osmosis, the dispersion medium move towards the anode.

(B) Diffused layer around colloidal particles in the electric double layer mostly contains negatively charged particles.

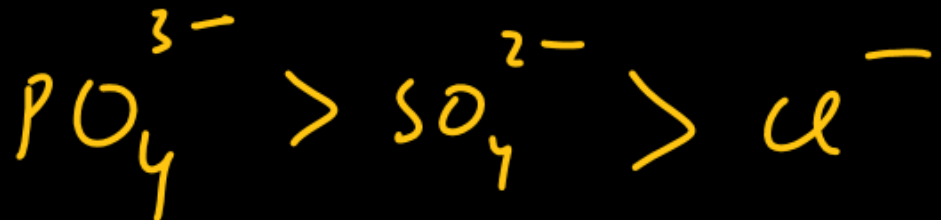
(C) It may be basic dye such as methylene blue.

(D) It may be metal sol.

→ +ve dye.

→ -ve sol

Solⁿ: $\text{Na}_3\text{PO}_4 > \text{CaSO}_4 > \text{AlCl}_3$. (From Hardy schulze rule)

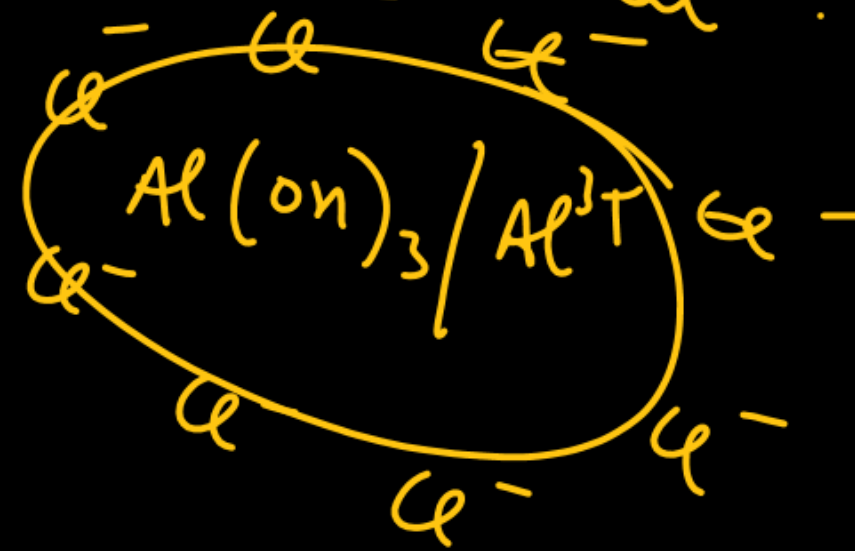


Anode : +ve terminal

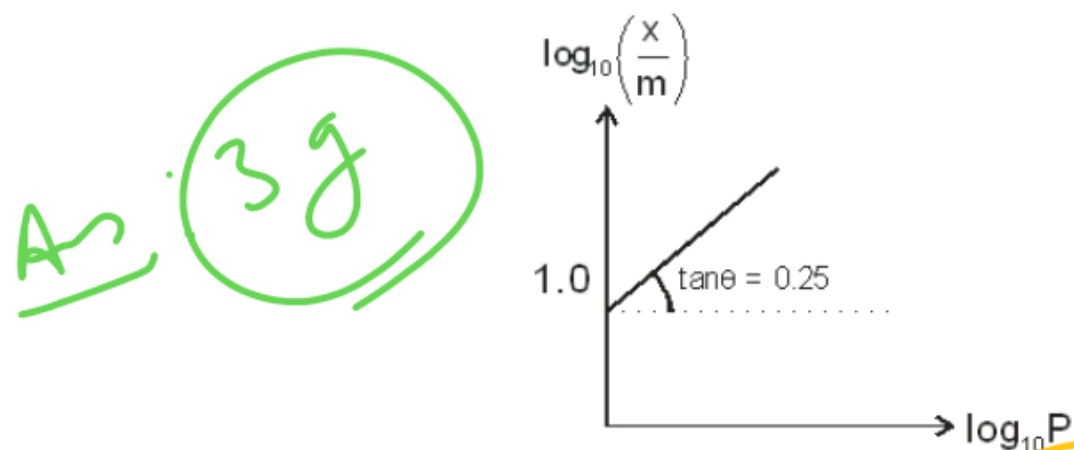
→ sol is +ve charged sol.

∴ dm is -ve charged

Ans: A, B, C.



Adsorption of hydrogen gas over palladium surface obey Freundlich's adsorption isotherm as shown in the graph. The mass (in g) of hydrogen gas adsorbed on the surface of 1.0 g palladium at a pressure of 8.1×10^{-3} atm is



Solⁿ:

$$\frac{x}{m} = K p^n \quad \Bigg| \quad \underbrace{\log \frac{x}{m}}_y = \log K + n \underbrace{\log p}_x$$

x = mass of adsorbent gas
 m = mass of solid adsorbent.

$$y = mx + c$$

$$\log K = 1$$

$$K = 10$$

$$n = 0.25$$

$$\frac{x_H}{1} = 10 \times (8.1 \times 10^{-3})^{1/4} \Rightarrow x = 10 \times (3)^{1/4} \times (10^{-1})^{1/4} \times 10^{1/4}$$

$$= \underline{3g} \text{ Ans.}$$

A quantity of 1.0 g of charcoal adsorbs 100 ml 0.5 M CH_3COOH to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.49 M. The surface area of the charcoal adsorbed by each molecule of acetic acid is (Surface area of charcoal = $3.0 \times 10^2 \text{ m}^2/\text{g}$) ($N_A = 6 \times 10^{23}$).

Solⁿ: (100 ml 0.5 M CH_3COOH (aq) $\longrightarrow$ 0.49 M, 100 ml. CH_3COOH .) adsorbed on 1 g of charcoal.

(Imp) $(100 \times 0.5 \times 10^{-3} - 100 \times 0.49 \times 10^{-3}) \times 6 \times 10^{23}$ molecules of CH_3COOH

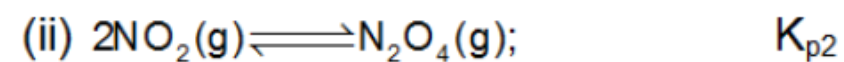
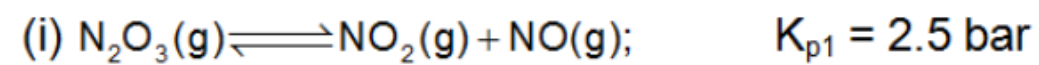
$\therefore$ 1 molecule of CH_3COOH ————— = $3 \times 10^2 \text{ m}^2$

$$= \frac{(50 - 49) \times 10^{-3} \times 6 \times 10^{23}}{1} \text{ m}^2$$

$$= \frac{1}{2} \times 10^{-18} \text{ m}^2$$

$$= 5 \times 10^{-19} \text{ m}^2/\text{molecule}$$

N_2O_3 is an unstable oxide of nitrogen and it decomposes into $NO(g)$ and $NO_2(g)$ where $NO_2(g)$ is further dimerised into N_2O_4 as

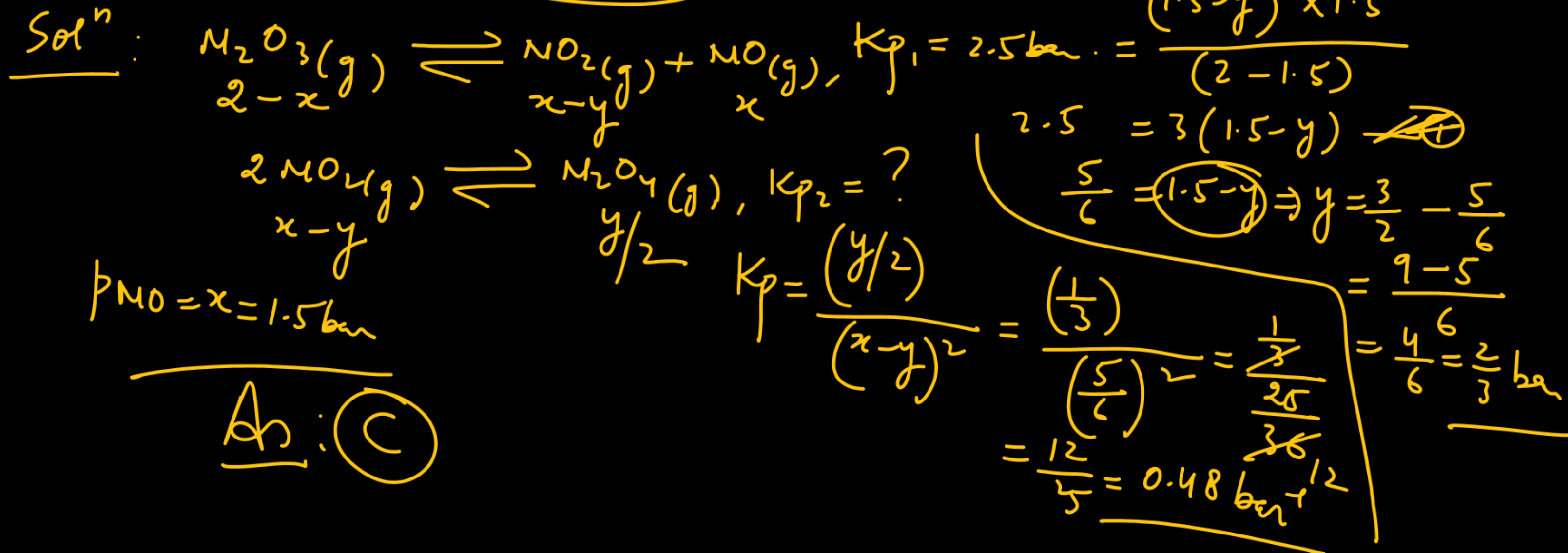


A flask is initially filled with pure $N_2O_3(g)$ having pressure 2 bar and equilibria were established. At equilibrium partial pressure of $NO(g)$ was found to be 1.5 bar.

The value of K_{p2} is :

- (A) 0.16 bar^{-1} (B) 0.32 bar^{-1} (C) 0.48 bar^{-1} (D) 0.64 bar^{-1}

← Concept of Simultaneous eqⁿ.

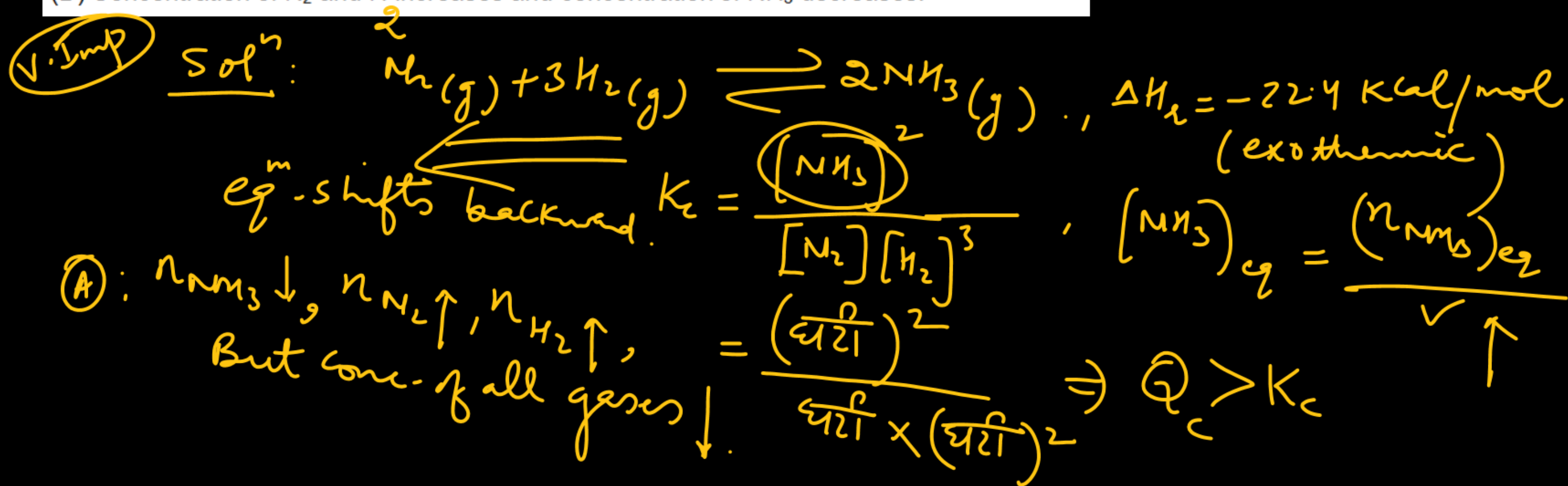


$\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ are allowed to react in a closed vessel at given temp. and pressure for the formation of $\text{NH}_3(\text{g})$ [$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + 22.4 \text{ kcal}$]

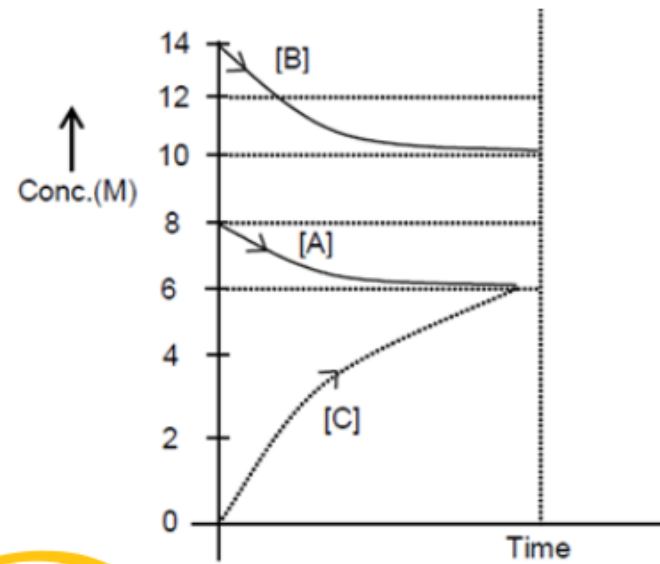
If $\text{He}(\text{g})$ is added at equilibrium at constant pressure then which is/are correct?

- ✓ (A) Concentration of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ and $\text{NH}_3(\text{g})$ decrease.
- ✓ (B) Moles of $\text{NH}_3(\text{g})$ decreases.
- ✓ (C) The extent of cooling depends on amount of $\text{He}(\text{g})$ added.
- (D) Concentration of N_2 and H increases and concentration of NH_3 decreases.

Ans: A, B, C

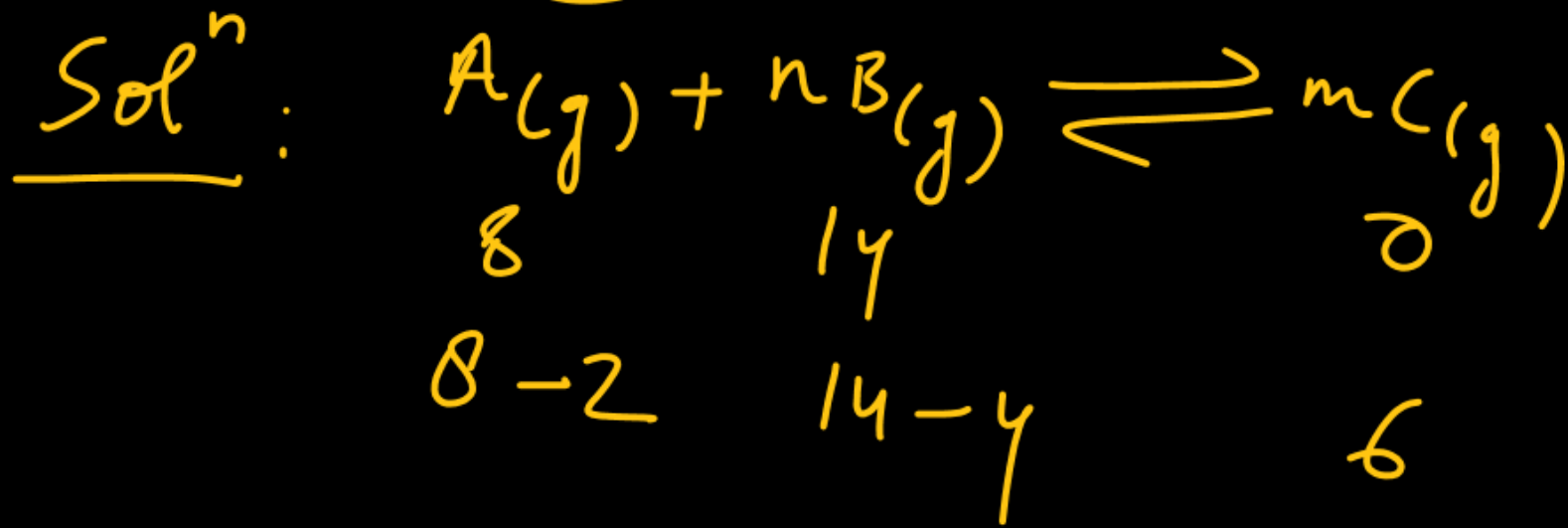


The gaseous reaction : $A(g) + nB(g) \rightleftharpoons mC(g)$ is represented by following curves



What is the value of $n + m$?

$= 5$



$$2n = 4$$

$$\underline{n = 2}$$

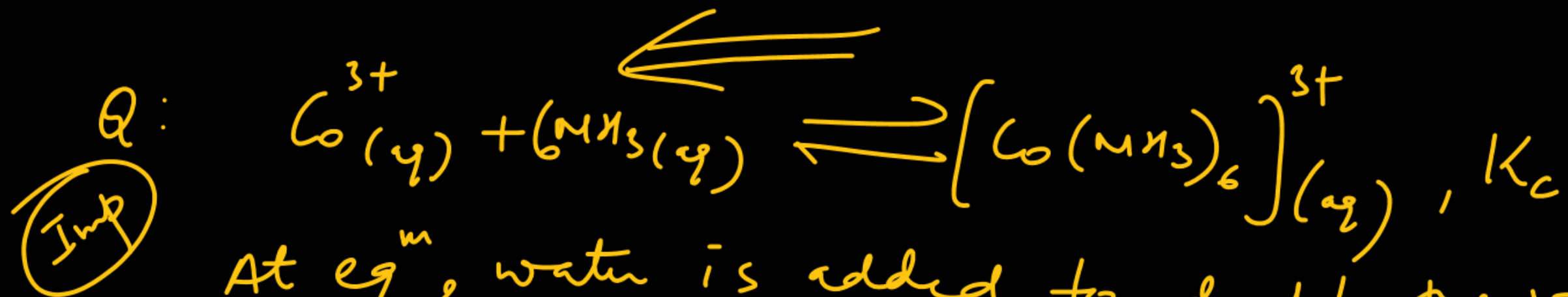
$$2m = 6$$

$$\underline{m = 3}$$

As: $n + m =$

$$2 + 3 = \underline{5}$$

✓



(Imp)

At eq^m , water is added to double the volume of system.

Then

(A) Eq^m shifts forward ~~(B) Eq^m shifts backward~~

(C) Conc. of Co^{3+} increases at new eq^m .

~~(D) Conc. of all species are decreased at new eq^m .~~

Solⁿ:

$$K_c = \frac{[\text{Co}(\text{NH}_3)_6]^{3+}}{[\text{Co}^{3+}][\text{NH}_3]^6}, \quad [\text{Co}^{3+}] = \frac{n_{\text{Co}^{3+}}}{V} \rightarrow 2V$$

$$= \frac{\text{घट}^1}{(\text{घट}^1)^7} \Rightarrow Q > K \text{ so } \text{eq}^m \text{ shifts backward.}$$

Ans: (B), (D)